Multiple Choice Questions (MCQ)

Solution 1: (c) 810 cm^3

Volume of the cuboid $= l \times b \times h = 15 \times 12 \times 4.5\ cm^3 = 810\ cm^3$

Solution 2: (b) 552 cm^2

Total surface area of the cuboid $= 2\ (lb + bh + lh)\ cm^2$

$= 2(12 \times 9 + 8 \times 9 + 12 \times 8)\ cm^2$

$= 2(108 + 72 + 96)\ cm^2$

$= 2 \times 276\ cm^2$

$= 552\ cm^2$

Solution 3: (b) 21 m^2

Length of the cuboid, $l = 15$ m

Breadth of the cuboid, $b = 6$ m

Height of the cuboid, $h = 5$ dm $= 0.5$ m $\quad (\because 1\ m = 10\ dm)$

$\therefore$ Lateral surface area of the cuboid $= 2h(l + b)$

$= 2 \times 0.5 \times (15 + 6)$

$= 2 \times 0.5 \times 21$

$= 21\ m^2$

Hence, the correct answer is option (b).

Solution 4: (c) 36 kg

Length $= 9$ m

Breadth $= 40$ cm $= 0.4$ m

Height $= 20$ cm $= 0.2$ m

$\therefore$ Weight of the beam = volume of the beam × weight of iron per cubic metre

$= 9 \times 0.4 \times 0.2 \times 50$

$= 36$ kg

Solution 5: (a) 15 m

Length of longest rod = diagonal of the room

= diagonal of a cuboid

$= \sqrt{l^2 + b^2 + h^2}$

$= \sqrt{10^2 + 10^2 + 5^2}$

$= \sqrt{100 + 100 + 25}\, m$

$= \sqrt{225}$ m

$= 15$ m

Solution 6: (d) 11.2 cm

Maximum length of the pencil = diagonal of the box

$= \sqrt{l^2 + b^2 + h^2}$

$= \sqrt{8^2 + 6^2 + 5^2}$

$= \sqrt{64 + 36 + 25}\, m$

$= \sqrt{125}$ m

$= 11.2$ m

Solution 7: (b) 192

$$\text{Number of planks} = \frac{\text{volume of the pit}}{\text{volume of 1 plank}} = \frac{40 \times 12 \times 16}{4 \times 5 \times 2} = \frac{7680}{40} = 192$$

Solution 8: (a) 480

Length of the pit = 20 m

Breadth of the pit = 6 m

Height of the pit = 50 cm = 0.5 m

Length of the plank = 5m

Breadth of the plank = 25 cm = 0.25 m

Height of the plank = 10 cm = 0.1 m

$$\therefore \text{Number of planks} = \frac{\text{volume of the pit}}{\text{volume of 1 plank}} = \frac{20 \times 6 \times 0.5}{5 \times 0.25 \times 0.1} = \frac{60 \times 1000}{125} = 480$$

Solution 9: (c) 6400

Length of the wall = 8 m = 800 cm

Breadth of the wall = 6 m = 600 cm

Height of the wall = 22.5 cm

Length of the brick = 25 cm

Breadth of the brick = 11.25 cm

Height of the brick = 6 cm

$$\therefore \text{Number of bricks required} = \frac{\text{volume of the wall}}{\text{volume of 1 brick}}$$

$$= \frac{800 \times 600 \times 22.5}{25 \times 11.25 \times 6}$$

$$= \frac{10800000}{1687.5}$$

$$= 6400$$

Solution 10: (b) 270

$$\text{Number of persons} = \frac{\text{volume of the hall}}{\text{volume of air required by 1 person}}$$

$$= \frac{20 \times 15 \times 4.5}{5}$$

$$= 20 \times 3 \times 4.5$$

$$= 270$$

$\therefore$ 270 persons can be accommodated.

Solution 11: (b) 2250 m^3

Length of the river $= 1.5$ m

Breadth of the river $= 30$ m

Depth of the river $= 3$ km $= 3000$ m

Now, volume of water that runs into the sea $= 1.5 \times 30 \times 3000$ m^3

$$= 135000 \text{ } m^3$$

$\therefore$ Volume of water that runs into the sea per minute $= \frac{135000}{60} = 2250 \text{ } m^3$

Solution 12: (d) 512 m^3

Suppose that 'a' m be the edge of the cube.

We have:

$4a^2 = 256$

$\Rightarrow a^2 = \frac{256}{4}$

$\Rightarrow a^2 = 64$

$\Rightarrow a = 8 \, m$

$\therefore$ Volume of the cube $= a^3 = 8^3 = 512 \text{ } m^3$

Solution 13: (c) 64 cm^3

Let 'a' cm be the edge of the cube.

We have:

$6a^2 = 96$

$\Rightarrow a^2 = \frac{96}{6}$

$\Rightarrow a^2 = 16$

$\Rightarrow a = 4\,m$

$\therefore$ Volume of the cube $= a^3 = 4^3 = 64\,cm^3$

Solution 14: (b) 384 cm^2

Suppose that a cm is the edge of the cube.

We have:

$a^3 = 512$

$\Rightarrow a = \sqrt[3]{512}$

$\Rightarrow a = 8\,cm$

$\therefore$ Total surface area of cube $= 6a^2 = 6 \times 8 \times 8 = 384\,cm^2$

Solution 15: (d) $10\sqrt{3}$ cm

Length of the longest rod = body diagonal of the vessel $= \sqrt{3}a = \sqrt{3} \times 10 = 10\sqrt{3}\,cm$

Solution 16: (b) 384 cm^2

We have:

length of diagonal of a cube $= 8\sqrt{3}$ cm

$\Rightarrow \sqrt{3}a = 8\sqrt{3}$

$\Rightarrow a = 8\,cm$

$\therefore$ Surface area of the cube $= 6a^2 = 6 \times 8 \times 8 = 384\,cm^2$

Solution 17: (d) 125%

Let a be the edge of the cube.

Then the surface area is $6a^2 = S$ (say)

Now, increased edge $= \left(a + \frac{50}{100}a\right) = \frac{150}{100}a = \frac{3}{2}a$

Then, new surface area $= 6\left(\frac{3}{2}a\right)^2$

$= 6 \times \frac{9}{4}a^2$

$= \frac{9}{4} \times 6a^2$

$= \frac{9}{4}S \quad [\because 6a^2 = S]$

Increase in surface area $= \frac{9}{4}S - S = \frac{5}{4}S$

$\therefore$ Percentage increase in surface area $= \frac{\frac{5}{4}S}{S} = 54 \times 100\% = 125\%$

Solution 18: (b) 144 cm^2

Volume of the new cube formed = total volume of the three cubes

Suppose that a cm is the edge of the new cube, then

$a^3 = 3^3 + 4^3 + 5^3$

$\Rightarrow a^3 = 27 + 64 + 125$

$\Rightarrow a^3 = 216$

$\Rightarrow a = 6$ cm

$\therefore$ Lateral surface area of the new cube $= 4a^2 = 4 \times 6 \times 6 = 144\ cm^2$

Solution 19: (d) 1000 m^3

Area of the land = 2 hectares $= 2 \times 10000 = 20000\ m^2$

Amount of rainfall $= 5$ cm $= 0.05$ m

$\therefore$ Volume of the water $=$ area of the land $\times$amount of rainfall

$= 2000 \times 0.05$

$= 1000\ m^3$

Solution 20: (c) 1 : 9

Suppose that the edges of the cubes are *a* and *b.*

We have:

$$\frac{a^3}{b^3} = \frac{1}{27}$$

$$\Rightarrow (ab)^3 = \frac{1}{27}$$

$$\Rightarrow ab = \frac{1}{3}$$

$$\therefore \text{Ratio of the surface areas} = \frac{6a^2}{6b^2} = \left(\frac{a}{b}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Solution 21: (d) becomes 8 times

Suppose that the side of the cube is *a.*

When it is doubled, it becomes $2a$.

New volume of the cube $= (2a)^3 = 8a^3$

Hence, the volume becomes 8 times the original volume.

Solution 22: (b) 396 cm^3

$$\text{Volume of the cylinder} = \pi r^2 h = \frac{22}{7} \times 3^2 \times 14 = 22 \times 9 \times 2 = 396\ cm^3$$

Solution 23: (b) 1760 cm^2

Curved surface area of the cylinder $= 2\pi rh$

$$= 2 \times \frac{22}{7} \times 14 \times 20$$

$$= 44 \times 40$$

$$= 1760\ cm^2$$

Solution 24: (c) 20 cm

Curved surface area $= 1760\ \text{cm}^2$

Suppose that h cm is the height of the cylinder.

Then we have:

$2\pi rh = 1760$

$\Rightarrow 2 \times \frac{22}{7} \times 14 \times h = 1760$

$\Rightarrow h = \frac{1760 \times 7}{44 \times 14}$

$\Rightarrow h = 20\ cm$

Solution 25: (b) 396 cm^3

Curved surface area $= 264\ \text{cm}^2$.
Let r cm be the radius of the cylinder.

Then we have:

$2\pi rh = 264$

$\Rightarrow 2 \times \frac{22}{7} \times r \times 14 = 264$

$\Rightarrow r = \frac{264 \times 7}{44 \times 14}$

$\Rightarrow r = 3\ cm$

$\therefore$ Volume of the cylinder $= \frac{22}{7} \times 3^2 \times 14 = 22 \times 9 \times 2 = 396\ \text{cm}^3$

Solution 26: (c) 6 m

Curved surface area $= 264\ m^2$

Volume $= 924\ m^3$

Let r m be the radius and h m be the height of the cylinder.

Then we have:

Curved surface area $= 2\pi rh$

$$\Rightarrow 264 = 2 \times \frac{22}{7} \times r \times h$$

$$\Rightarrow 264 = \frac{44}{7} \times r \times h$$

$$\Rightarrow r = \frac{264 \times 7}{44 \times h}$$

$$\Rightarrow r = \frac{42}{h}$$

Volume $= \pi r^2 h$

$$\Rightarrow 924 = \frac{22}{7} \times \frac{42}{h} \times \frac{42}{h} \times h$$

$$\Rightarrow 924 = 22 \times \frac{6}{h} \times 42$$

$$\Rightarrow \frac{924}{22 \times 42 \times 6} = \frac{1}{h}$$

$$\Rightarrow \frac{924}{5544} = \frac{1}{h}$$

$$\Rightarrow h = \frac{5544}{924}$$

$$\Rightarrow h = 6\ m$$

Solution 27: (c) 10 : 9

Suppose that the radii of the cylinders are 2r and 3r and their respective heights are 5h and 3h.

Then, ratio of the curved surface areas $= \frac{2\pi \times 2r \times 5h}{2\pi \times 3r \times 3h} = \frac{10}{9} = 10 : 9$

Solution 28: (b) 20 : 27

Suppose that the radii of the cylinders are $2r$ and $3r$ and their respective heights are $5h$ and $3h$..

$\therefore$ Ratio of their volumes $= \dfrac{\pi \times (2r)^2 \times 5h}{\pi \times (3r)^2 \times 3h} = \dfrac{4r^2 \times 5}{9r^2 \times 3} = \dfrac{20}{27} = 20 : 27$

Solution 29: (d) 770 cm^2

We have:

$r : h = 2 : 3$

$\Rightarrow \dfrac{r}{h} = \dfrac{2}{3}$

$\Rightarrow h = \dfrac{3}{2}r$

Now, volume $= 1617\ cm^3$

$\Rightarrow \pi r^2 h = 1617$

$\Rightarrow \dfrac{22}{7} \times r^2 \times \dfrac{3}{2}r = 1617$

$\Rightarrow r^3 = \dfrac{1617 \times 14}{66}$

$\Rightarrow r^3 = 343$

$\Rightarrow r = 7\ cm$

We have $h = 10.5$ cm

Hence, total surface area $= 2\pi rh + 2\pi r^2$

$= 2\pi r(h + r)$

$= 2 \times \dfrac{22}{7} \times 7(10.5 + 7)$

$= 44 \times 17.5$

$= 770\ cm^2$

Solution 30: (b) $\sqrt{2}:1$

Suppose that the heights of two cylinders are h and $2h$ whose radii are r and R, respectively.

Since the volumes of the cylinders are equal, we have:

$\pi r^2 h = \pi R^2 \times 2$

$\Rightarrow \frac{r^2}{R^2} = 21$

$\Rightarrow \left(\frac{r}{R}\right) = \sqrt{\frac{2}{1}}$

$\Rightarrow r: R = \sqrt{2}:1$

Solution 31: (a) 1078 cm^3

We have:

The ratio between the curved surface area and total surface is given by,

$\frac{2\pi rh}{2\pi rh + 2\pi r^2} = \frac{1}{2}$

$\Rightarrow \frac{2\pi rh}{2\pi r(h+r)} = \frac{1}{2}$

$\Rightarrow \frac{h}{(h+r)} = \frac{1}{2}$

$\Rightarrow 2h = h + r$

$\Rightarrow h = r$

Now,

Total Surface area $= 616$

$\Rightarrow 2\pi rh + 2\pi r^2 = 616$

$\Rightarrow 2\pi r(h+r) = 616$

$\Rightarrow 2\pi(r+r) = 616 \qquad [\because h = r]$

$\Rightarrow 2\pi \times 2r = 616$

$\Rightarrow 4\pi r = 616$

$$\Rightarrow 4 \times \frac{22}{7} \times r^2 = 616$$

$$\Rightarrow r^2 = \frac{616 \times 8}{4 \times 22}$$

$$\Rightarrow r^2 = 49$$

$$\Rightarrow r = 9\ cm$$

$$\therefore h = 9\ cm$$

So, Volume of the cylinder $= \pi r^2 h = \frac{22}{7} \times 7^2 \times 7 = 22 \times 49 = 1078\ cm^3$

Solution 32: (c) halved

Suppose that the new radius is $\frac{1}{2}r$ and the height is $2h$.

$$\therefore \text{Volume} = \pi \times \left(\frac{1}{2}r\right)^2 \times 2h = \pi \times \frac{r^2}{4} \times 2h = \frac{1}{2}\pi r^2 h$$

Solution 33: (b) 450

$$\text{Number of coins} = \frac{\text{volume of cylinder to be formed}}{\text{volume of a coin}}$$

$$= \frac{\pi \times 2.25 \times 2.25 \times 10}{\pi \times 0.75 \times 0.75 \times 0.2}$$

$$= \frac{225 \times 225 \times 50}{75 \times 75}$$

$$= 450$$

Solution 34: (d) 9 times

Let r and h be the original radius and length of the wire

$\therefore$ New radius be $\frac{1}{3}$r.

Let the length is H.

The volume remains the same.

$i.e., \pi r^2 l = \pi \times \left(\frac{1}{3}r\right)^2 \times H$

$\Rightarrow h = 19H$

$\Rightarrow H = 9h$

$\therefore$ The new length becomes nine times the original length.

Solution 35: (b) 1320 m^2

Area covered by the roller in 1 revolution $= 2\pi rh$

$= 2 \times \frac{22}{7} \times 42 \times 100$

$= 44 \times 600$

$= 26400\ cm^2$

$\therefore$ Area covered in 50 revolutions $= 26400 \times 500\ cm^2 = 1320\ m^2$

Solution 36: (b) 112 m

Volume of the lead = volume of the cylindrical wire

Suppose that h m is the length of the wire.

As 2.2 dm^3 of lead is to be drawn into a cylindrical wire of diameter 0.50 cm, we have:

Now, 2.2 $dm^3 = (2 \times 1000) = 2200$ cm

radius, $r = \frac{0.50}{2} = 0.25\ cm$

$\therefore$ Volume of the cylindrical wire = 2200 cm

$\Rightarrow \pi r^2 h = 2200$

$\Rightarrow \frac{22}{7} \times (0.25)^2 \times h = 2200$

$\Rightarrow \frac{22}{7} \times 0.0625 \times h = 2200$

$\Rightarrow h = \frac{2200 \times 7}{22 \times 0.0625}$

$\Rightarrow h = \frac{700}{0.0625}$

$\Rightarrow h = 11200\ cm$

$\Rightarrow h = 112\ m$

Solution 37: (c) $2\pi rh$

The lateral surface area of a cylinder is equal to its curved surface area.

$\therefore$ Lateral surface area of a cylinder $= 2\pi rh$

Solution 38: (b) 550 cm^2

The height of a cone is 24 cm and the diameter of its base is 14 cm

So, it's radius = 7 cm

Now, we know that

$l = \sqrt{r^2 + h^2} = \sqrt{7^2 + (24)^2} = \sqrt{49 + 576} = 25\ cm$

$\therefore$ Curved surface area of the cone $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550\ cm^2$

Solution 39: (d) $(144\pi)\ cm^3$

Volume of the cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 6^2 \times 12 = \pi \times 36 \times 4 = 144\pi\ cm^3$

Solution 40: (c) 220 m

Let the length of the required cloth be L m and its breadth be B m.

Here, B = 2.5 m

Radius of the conical tent, r =7 m

Height of the tent, h = 24 m

Area of cloth required = curved surface area of the conical tent

$\Rightarrow L \times B = \pi r l$

$\Rightarrow L \times 2.5 = \frac{22}{7} \times 7 \times \sqrt{r^2 + h^2}$

$\Rightarrow 2.5L = 22 \times \sqrt{(7)^2 + (24)^2}$

$\Rightarrow 2.5L = 22 \times \sqrt{49 + 576}$

$\Rightarrow 2.5L = 22 \times \sqrt{625}$

$\Rightarrow 2.5L = 22 \times 25$

$\Rightarrow L = \frac{550}{2.5}$

$\Rightarrow L = 220\ m$

Solution 41: (a) 10 cm

Let r cm be the radius of the cone.

Volume $= 1570\ cm^3$

Then,

$\frac{1}{3} \times 3.14 \times r^2 \times 15 = 1570$

$3.14 \times r^2 \times 5 = 1570$

$\Rightarrow r^2 = \frac{1570}{3.14 \times 5}$

$\Rightarrow r^2 = 100$

$\Rightarrow r = 10\ cm$

Solution 42: (b) 7546 cm^3

Radius of the cone, $r = \sqrt{l^2 - h^2} = \sqrt{(28)^2 - (21)^2} = \sqrt{784 - 441} = \sqrt{343}\ cm$

$\therefore$ Volume of the cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times \left(\sqrt{343}\right)^2 \times 21 = 22 \times 343 = 7546\ cm^3$

Solution 43: (c) 550 cm^2

Let r cm be the radius of the cone.

Volume of the right circular cone $= 1232\ \text{cm}^3$

Then we have:

$$\frac{1}{3} \times \frac{22}{7} \times r^2 \times 24 = 1232$$

$$\Rightarrow r^2 = \frac{1232 \times 21}{22 \times 24}$$

$$\Rightarrow r^2 = 49$$

$$\Rightarrow r = 7\ cm$$

$\therefore$ Curved surface area of the cone $= \pi rl$

$$= \frac{22}{7} \times 7 \times \sqrt{7^2 + (24)^2}$$

$$= 22 \times \sqrt{49 + 576}$$

$$= 22 \times \sqrt{625}$$

$$= 22 \times 25$$

$$= 550\ \text{cm}^2$$

Solution 44: (d) 25 : 64

Suppose that the radii of the cones are $4r$ and $5r$ and their heights are h and H, respectively .

It is given that the ratio of the volumes of the two cones is 1 : 4

Then we have:

$$\frac{\frac{1}{3}\pi(4r)^2 h}{\frac{1}{3}\pi(5r)^2 H} = \frac{1}{4}$$

$$\Rightarrow \frac{16r^2 h}{25r^2 H} = \frac{1}{4}$$

$$\Rightarrow \frac{16h}{25H} = \frac{1}{4}$$

$\Rightarrow \frac{h}{H} = \frac{25}{64}$

$\Rightarrow h : H = 25 : 64$

Solution 45: (a) 100 %

Suppose that height of the cone becomes $2h$ and let its radius be r.

Then new volume of the cone $= \frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2 h = 2 \times$ volume of the cone

Increase in volume $= \frac{2}{3}\pi r^2 h - \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r 2h$

$\therefore$ Percentage increase $=$ increase in volume initial volume $\times 100$ %

$$= \frac{\frac{1}{3}\pi r^2 h}{\frac{1}{3}\pi r^2 h} \times 100\%$$

$$= 100\%$$

Hence, the volume increases by 100%.

Solution 46: (b) 4 : 1

If the slant height of the first cone is l, then the slant height of the second cone will be $2l$.

Let the radii of the first and second cones be r and R, respectively.

Then we have:

$\pi r l = 2 \times (\pi R \times 2l)$

$\Rightarrow r = 4R$

$\Rightarrow \frac{r}{R} = \frac{4}{1}$

Solution 47: (b) 3 : 1

It is given that the right circular cylinder and the right circular cone have the same base and height.

Suppose that the respective radii of bases and heights are equal to r and h.

Then ratio of their volumes $= \frac{\pi r^2 h}{\frac{1}{3}\pi r^2 h} = \frac{3}{1} = 3:1$

Solution 48: (d) 1 : 3

It is given that the right circular cylinder and the right circular cone have the same radius and volume.

Suppose that the radii of their bases are equal to r and the respective heights of the cylinder and the cone are h and H.

As the volumes of the cylinder and the cone are the same, we have:

$\pi r^2 h = \frac{1}{3}\pi r^2 H$

$\Rightarrow \frac{h}{H} = \frac{1}{3}$

Solution 49: (a) 9 : 8

Suppose that the respective radii of the cylinder and the cone are $3r$ and $4r$ and their respective heights are $2h$ and $3h$.

$\therefore$ Ratio of the volumes $= \frac{\pi(3r)^2 \times 2h}{\frac{1}{3}\pi(4r)^2 \times 3h} = \frac{3 \times 9 \times 2}{16 \times 3} = \frac{9}{8}$

Solution 50: (d) 8 times

Let the new height and radius be $2h$ and $2r$, respectively.

New volume of the cone $= \frac{1}{3}\pi(2r)^2 \times 2h = \frac{8}{3}\pi r^2 h = 8 \times$ initial volume of the cone

Solution 51: (d) 13500

Radius of the cylinder $= 3$ cm

Height of the cylinder $= 5$ cm

Radius of the cone $= 1$ mm $= 0.1$ cm

Height of the cone $= 1$ cm

$$\therefore \text{ Number of cones, } n = \frac{\text{volume of the cylinder}}{\text{volume of 1 cone}}$$

$$= \frac{\pi \times 3^2 \times 5}{\frac{1}{3}\pi \times (0.1)^2 \times 1}$$

$$= \frac{3 \times 9 \times 5}{0.01}$$

$$= 13500$$

Solution 52: (b) 15 m

Suppose that the height of the cone is h m.

Area of the ground $= 11 \times 4 = 44$ m^2

$$\therefore \pi r^2 = 44$$

$$\Rightarrow r^2 = \frac{44 \times 7}{22}$$

$$\Rightarrow r^2 = 14$$

Also,

$$13\pi r^2 h = 220$$

$$\Rightarrow \frac{1}{3} \times \frac{22}{7} \times 14h = 220$$

$$\Rightarrow h = \frac{220 \times 21}{22 \times 14}$$

$$\Rightarrow h = 15\, m$$

Hence, the height of the cone is 15 m.

Solution 53: (a) $\frac{32\pi r^3}{3}$

Volume of the sphere of radius $2r = \frac{4}{3}\pi(2r)^3 = \frac{4}{3}\pi 8r^3 = \frac{32}{3}\pi r^3$

Solution 54: (b) 4851 cm^3

Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 10.5$

$= 88 \times 0.5 \times 10.5 \times 10.5$

$= 4851\ cm^3$

Solution 55: (d) 5544 cm^2

Surface area of sphere $= 4\pi r^2 = 4 \times \frac{22}{7} \times 21 \times 21 = 5544\ cm^2$

Solution 56: (c) 4851 cm^3

Surface area $= 1386\ cm^2$

Let r cm be the radius of the sphere.

Then we have:

$4\pi r^2 = 1386$

$\Rightarrow r^2 = \frac{1386 \times 7}{4 \times 22}$

$\Rightarrow r^2 = 110.25$

$\Rightarrow r = 10.5\ cm$

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 10.5$

$= 4851 \text{ cm}^3$

Solution 57: (a) (288π) m^3

Surface area $= (144\pi)$ m^2

Ler rm be the radius of the sphere.

Then we have:

$4\pi r^2 = 144\pi$

$\Rightarrow r^2 = \frac{144}{4}$

$\Rightarrow r^2 = 36$

$\Rightarrow r = 6\ m$

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3}\pi \times 6 \times 6 \times 6$

$= 288\pi \text{ m}^3$

Solution 58: (a) 5544 cm^2

Let $r$cm be the radius of the sphere.

Then we have:

$\frac{4}{3}\pi r^3 = 38808$

$\Rightarrow r^3 = \frac{38808 \times 3 \times 7}{4 \times 22}$

$\Rightarrow r^3 = 9261$

$\Rightarrow r = 21\ cm$

$\therefore$ Curved surface area $= 4\pi r^2 = 4 \times \frac{22}{7} \times 21 \times 21 = 5544\ \text{cm}^3$

Solution 59: (b) 1 : 4

Suppose that r and R are the radii of the spheres.

Then we have:

$$\frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = \frac{1}{8}$$

$$\Rightarrow \left(\frac{r}{R}\right)^3 = \left(\frac{1}{2}\right)^3$$

$$\Rightarrow \frac{r}{R} = \frac{1}{2}$$

$\therefore$ Ratio of surface area of spheres $= \frac{4\pi r^2}{4\pi R^2} = \left(\frac{r}{R}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} = 1 : 4$

Solution 60: (d) 64

$$\text{Number of balls} = \frac{\text{volume of solid metal ball}}{\text{volumeof 1 small ball}}$$

$$= \frac{\frac{4}{3}\pi \times 8^3}{\frac{4}{3}\pi \times 2^3}$$

$$= \frac{512}{8}$$

$$= 64$$

Solution 61: (b) 2.1 cm

Let r cm be the radius of the sphere.

Volume of the cone = volume of the sphere

$$\Rightarrow \frac{1}{3}\pi \times 2.1 \times 2.1 \times 8.4 = \frac{4}{3}\pi r^3$$

$$\Rightarrow r^3 = 2.1 \times 2.1 \times 2.1$$

$$\Rightarrow r = 2.1\ cm$$

Solution 62: (b) 288 m

Let h m be the length of wire.

Volume of the lead ball = volume of the wire

$$\Rightarrow \frac{4}{3}\pi \times 6^3 = \pi \times (0.1)^2 h$$

$$\Rightarrow h = \frac{4 \times 216}{3 \times 0.01}$$

$$\Rightarrow h = 28800\ cm$$

$$\Rightarrow h = 288\ m$$

Solution 63: (c) 126

$$\text{Number of cones} = \frac{\text{volume of the sphere}}{\text{volume of 1 cone}}$$

$$= \frac{\frac{4}{3}\pi \times (10.5)^3}{\frac{1}{3}\pi \times (3.5)^2 \times 3}$$

$$= \frac{4 \times (10.5)^3}{3 \times (3.5)^2}$$

$$= 126$$

Solution 64: (d) 84000

$$\text{Number of lead shots} = \frac{\text{volume of cuboid}}{\text{volume of 1 lead shot}}$$

$$= \frac{9 \times 11 \times 12}{\frac{4}{3} \times \frac{22}{7} \times \left(\frac{0.3}{2}\right)^3}$$

$$= \frac{9 \times 11 \times 3 \times 3 \times 7}{22 \times 0.15 \times 0.15 \times 0.15}$$

$= 84000$

Solution 65: (c) 36 m

Let h m be the length of the wire.

Volume of the sphere = volume of the wire

$\Rightarrow \frac{4}{3}\pi \times 3^3 = \pi \times (0.1)^2 \times h$

$\Rightarrow 4 \times 9 = (0.1)^2 \times h$

$\Rightarrow 36 = (0.1)^2 \times h$

$\Rightarrow h = \frac{36}{(0.1)^2}$

$\Rightarrow h = \frac{36}{0.001}$

$\Rightarrow h = 3600\ cm$

$\Rightarrow h = 36\ m$

Solution 66: (a) 6.3 cm

Let r cm be the radius of base of the cone.

Volume of the sphere = volume of the cone

$\Rightarrow \frac{4}{3}\pi \times (6.3)^3 = \frac{1}{3}\pi \times r^2 \times 25.2$

$\Rightarrow 4 \times (6.3)^3 = r^2 \times 25.2$

$\Rightarrow r^2 = \frac{4 \times 6.3 \times 6.3 \times 6.3}{25.2}$

$\Rightarrow r^2 = 39.69$

$\Rightarrow r^2 = \sqrt{39.69}$

$\Rightarrow r = 6.3\ cm$

Solution 67: (c) 2.5 cm

Let $r$cm be the radius of the third ball.

Volume of the original ball = volume of the three balls

$$\Rightarrow \frac{4}{3}\pi \times 3^3 = \frac{4}{3}\pi \times (1.5)^3 + \frac{4}{3}\pi \times 2^3 + \frac{4}{3}\pi r^3$$

$$\Rightarrow \frac{4}{3}\pi \times 3^3 = \frac{4}{3}\pi[(1.5)^3 + 2^3 + r^3]$$

$$\Rightarrow 3^3 = (1.5)^3 + 2^3 + r^3$$

$$\Rightarrow 27 = 3.375 + 8 + r^3$$

$$\Rightarrow r^3 = 27 - 11.375$$

$$\Rightarrow r^3 = 15.625$$

$$\Rightarrow r = 2.5\ cm$$

Solution 68: (a) 1 : 4

Ratio of the surface areas of balloon $= \frac{2\pi \times 6^2}{2\pi \times 12^2} = \frac{36}{144} = \frac{1}{4}$

Solution 69: (d) 88 cm^2

Let the radii be r cm and (7-r) cm

Volume of the two spheres are in the ratio 64:27

$$\Rightarrow \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi(7-r)^3} = \frac{64}{27}$$

$$\Rightarrow \left(\frac{r}{7-r}\right)^3 = \left(\frac{4}{3}\right)^3$$

$$\Rightarrow \frac{r}{7-r} = \frac{4}{3}$$

$$\Rightarrow 3r = 28 - 4r$$

$$\Rightarrow 7r = 28$$

$$\Rightarrow r = 4\ cm$$

So, their radii are 4cm and (7-r) = 3cm

Now, Difference of their total surface areas

$$= 4\pi(4)^2 - 4\pi(3)^2$$

$= 4 \times \frac{22}{7}(16 - 9)$

$= 88\ cm^2$

Solution 70: (c) 54

$$\text{Number of bottles} = \frac{\text{volume of hemispherical bowl}}{\text{volume of 1 bottle}}$$

$$= \frac{\frac{2}{3} \times \pi \times 9^3}{\pi \times (1.5)^2 \times 4}$$

$$= \frac{2 \times 729}{2.25 \times 12}$$

$= 54$

Solution 71: (b) 2 : 1

Let the radii of the cone and the hemisphere be r and their respective heights be h and H.

Then

$$\frac{1}{3}\pi \times r^2 \times h = \frac{2}{3}\pi \times r^2 \times H$$

$$\Rightarrow h = 2H$$

$$\Rightarrow \frac{h}{H} = \frac{2}{1}$$

$$\Rightarrow \frac{h}{H} = 2 : 1$$

Solution 72: (a) 1 : 2 : 3

The cone, hemisphere and the cylinder stand on equal bases and have the same height.

We know that radius and height of a hemisphere are the same.

Hence, the height of the cone and the cylinder will be the common radius.

i.e., $r = h$

Ratio of the volumes of the cone, hemisphere and the cylinder:

$\frac{1}{3}\pi r^2 : \frac{2}{3}\pi r^3 : \pi r^2(h)$

$\Rightarrow \frac{1}{3}\pi r^2 : \frac{2}{3}\pi r^3 : \pi r^2(r) \qquad [\because r = h]$

$\Rightarrow \frac{1}{3} : \frac{2}{3} : 1 \qquad [\because r = h]$

$\Rightarrow 1 : 2 : 3$

Solution 73: (c) 3 units

We have:

$\frac{4}{3}\pi r^3 = 4\pi r^2$

$\Rightarrow \frac{1}{3} r = 1$

$\Rightarrow r = 3$ units